Python Programming Techniques
TIU-UCA-T315

L-T-P: 2-1-0 **Credits: 3**

COURSE OBJECTIVES

The aim of this course is to

- To acquire programming skills in core Python.
- To acquire Object Oriented Skills in Python.
- To develop the skill of designing Graphical user Interfaces in Python.
- To develop the ability to write database applications in Python

Detailed Topics:

Introduction to Python , The basic elements of python, Branching Programs, Control Structures, Strings and Input, Iteration, Functions, Scoping and Abstraction Functions and scoping, Specifications, Recursion, Global variables, Modules, Files, System Functions and Parameters, Structured Types, Mutability and Higher-Order Functions Strings, Tuples, Lists and Dictionaries, Lists and Mutability, Functions as Objects, Testing, Debugging, Exceptions and Assertions, Types of testing – Black-box and Glass-box, Debugging, Handling Exceptions, Assertions, Classes and Object-Oriented Programming Abstract Data Types and Classes, Inheritance, Encapsulation and Information Hiding, Simple Algorithms and Data structures Search Algorithms, Sorting Algorithms, Hash Tables, Regular Expressions – REs and Python, Plotting using PyLab.

Books for Main Reading:

1. John V Guttag. "Introduction to Computation and Programming Using Python", Prentice Hall of India
2. R. Nageswara Rao, "Core Python Programming", dreamtech
3. Wesley J. Chun. "Core Python Programming - Second Edition", Prentice Hall
4. Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser, "Data Structures and Algorithms in Pyhon", Wiley.

Books for Supplementary Reading:

1. Kenneth A. Lambert, "Fundamentals of Python – First Programs", CENGAGE Publication
2. Luke Sneeringer, "Professional Python", Wrox
3. "Hacking Secret Ciphers with Python", Al Sweigart,
URLhttps://inventwithpython.com/hacking/chapters

COURSE OUTCOME:

After completion of this course students will be able to:

CO1: Describe the Numbers, Math functions, Strings, List, Tuples and Dictionaries inPython
CO2: Express different Decision Making statements and Functions
CO3: Interpret Object oriented programming in Python
CO4: Understand and summarize different File handling operations
CO5: Explain how to design GUI Applications in Python and evaluate different database operations
CO6: Design and develop Client Server network applications using Python